

TABLE II
WAVEFORMS VIEWERS

Name	Developer	Platform	Wave formats
Styx	Norman Hendrich	Unix	Only Hades file formats
Dinotrace	Wilson Snyder and Allen Gallotta	Linux, Solaris, Windows	.vcd, .ascii
EPWave	Victor EDA	Web browser based	.vcd



Fig. 2: Demonstrating application integration with Mac OS/X/Quartz



Fig. 3: GTKWave running under Windows

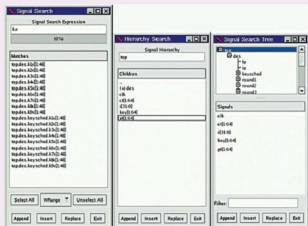


Fig. 4: Types of searches

for remote control by other applications. This can be enabled because of the availability of TCL scripting and call-back capability.

Another important feature is called TwinWave, which is a GTKPlug helping you manage two or more sessions of GTKWave in either a single window or in two separate windows. Scrolling and other GUI manipulations can be performed in lockstep across both sessions.

Traditionally, in integrated circuit design, waveform viewers are typically used in conjunction with a simulator. Not only does this help in individual assessment of signals but it also shows the effect of signals interacting with each other in the circuit. In the industry, though, operation is not as dynamic, as engineers run batches of simulations including all possible sets before running the waveforms to check for consistency. Value change dump (VCD) is the most preferred format for saving simulation dump files, while some companies have their own proprietary formats that include extra details depending on the developer.

Techniques like named pipes and shared memory proxies are used to facilitate the same. It allows you to navigate the dump as it is being written to the pipe and watch the simulation in real time. GTKPlug mechanism further allows you to integrate the viewer with other simulators in order to provide an interactive environment, all in one window.

Curiously enough, GTKWave is the only open source waveform viewer that supports post-mortem viewing as well as interactive viewing of VCD files. Presence of this feature allows you to write the output of a simulator and then feed the same to the waveform viewer. **EFY**

current hierarchy as well as the children box, which contains all signals in the immediate hierarchy below the searched parent term. In this format, in order to append, replace or insert, you need to select individual ones, otherwise the selected parent and children will be acted upon.

Tree search. This is the most used search requester, as it can be directly embedded onto GTKWave versions 2.4 and above. It has a top tree selection box, signals box and POSIX regular expression filter. The filter allows you to filter a specific set of signals that you might want to investigate further. If, for example, you are working on a synthesised netlist, then you will find that this tree-searching feature helps you filter signal names at the highest level, saving you a lot of time.

Why you should go for GTKWave

GTKWave has provisions in its code

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